THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 10: **How Quantities Combine (Understanding Data)** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. 660 mL of a 13% sugar solution is mixed with 440 mL of a 38% sugar solution. The concentration of the mixture is: 1

(a) 24% **(b)** 22% **(c)** 25% **(d)** 23%

2. Marks of 76, 63 and 54 are given weights 6, 8 and 3 respectively. The weighted average is: 1

(a) 65 **(b)** 66 **(c)** 67 **(d)** 68

3. The average monthly salary of 19 employees is ₹52 thousand. An employee earning ₹39 thousand leaves and a new employee joins, and the average rises by ₹2 thousand. The new employee's salary (in ₹ thousand) is: 1

(a) 77 **(b)** 79 **(c)** 75 **(d)** 82

4. A shop earns a profit of 21% on sales of ₹6000 and 26% on sales of ₹9000. The overall profit percentage on total sales is: 1

(a) 25% **(b)** 26% **(c)** 23% **(d)** 24%

5. The average of 30 numbers is 81. When a number 50 is included, the average of all 31 numbers becomes: 1

(a) 79 **(b)** 80 **(c)** 81 **(d)** 82

6. The average of 15 numbers is 78. If the number 50 is removed, the average of the remaining 14 numbers is: 1

(a) 81 **(b)** 82 **(c)** 79 **(d)** 80

7. The average of 40 numbers is 64. The average of all 62 numbers (including 22 more numbers) is 42. The average of the 22 additional numbers is: 1

(a) 1 **(b)** 4 **(c)** 3 **(d)** 2

8. A group of 36 people has average weight 40 kg. Another group has average weight 51 kg. Together the average weight is 49 kg. The number of people in the second group is: 1

(a) 162 **(b)** 163 **(c)** 164 **(d)** 161

9. 400 mL of a 59% sugar solution is mixed with 800 mL of a 11% sugar solution. The concentration of the mixture is: 1

(a) 29% **(b)** 28% **(c)** 27% **(d)** 26%

10. Marks of 77, 89 and 79 are given weights 4, 4 and 8 respectively. The weighted average is: 1

(a) 83 **(b)** 80 **(c)** 81 **(d)** 82

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A cyclist rides from home to a market at 30 km/h and returns by the same route at 42 km/h. Find the average speed for the whole trip. 2

12. The mean of 44 observations is 52. One observation 138 is removed. Find the mean of the remaining observations. 2

13. The average monthly salary of 20 employees is ₹69 thousand. An employee earning ₹23 thousand leaves and a new employee joins; the average rises by ₹4 thousand. Find the new employee's monthly salary. 2

14. A cyclist rides from home to a market at 40 km/h and returns by the same route at 60 km/h. Find the average speed for the whole trip. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In a class, 38 students have an average of 40 marks and the remaining students have an average of 55 marks. The class average is 45. Find the number of remaining students. 3

16. The average score of 40 students in a test is 87. When 36 more students take the test, the average score of all 76 students becomes 60. Find the average score of the 36 additional students. 3

17. 130 mL of a 19% acid solution is diluted by adding 60 mL of water. Find the new concentration. 3

OR

In a class, 31 students have an average of 43 marks and the remaining students have an average of 52 marks. The class average is 49. Find the number of remaining students. 3

18. The average score of 25 students in a test is 74. When 50 more students take the test, the average score of all 75 students becomes 64. Find the average score of the 50 additional students. 3

OR

50 mL of a 40% acid solution is diluted by adding 50 mL of water. Find the new concentration. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Three sections have 45, 48 and 37 students with average marks 74, 57 and 82 respectively. Find the average marks of all the students. 5

OR

300 mL of a 48% sugar solution is mixed with 750 mL of a 34% sugar solution. Find the percentage concentration of the mixture. 5

20. Three sections have 36, 47 and 47 students with average marks 69, 50 and 88 respectively. Find the average marks of all the students. 5

OR

800 mL of a 39% sugar solution is mixed with 550 mL of a 12% sugar solution. Find the percentage concentration of the mixture. 5

END OF QUESTION PAPER — Total: 40 Marks